The Constitutional Trilemma: Aggregation, Strategy, and Truth in Constitutional AI

A 2026 Anthropic Fellowship Research Proposal — Economics Workstream

Abstract. I prove an impossibility theorem: any multi-critic Constitutional AI pipeline (Bai et al. 2022, arXiv:2212.08073) must give up at least one of aggregation rationality, strategic robustness, or informational truthfulness. I then propose an empirical program to quantify the price-of-pluralism this trilemma imposes on Anthropic's flagship alignment stack.

1. Research Question. Can the constitutional-AI pipeline jointly satisfy (a) aggregation rationality across plural principles, (b) strategic robustness against paraphrase attacks and prompt-engineering, and (c) informational truthfulness when the operator-held constitution prior diverges from the deployment-time prior? Conitzer, Freedman, Heitzig, Holliday, Kristensen, Procaccia, and Russell (2024, arXiv:2404.10271) make the case that social choice should guide AI alignment. Ge, Tang, Procaccia, and Zampetakis (NeurIPS 2024, arXiv:2405.14758) supply axiomatic foundations for RLHF. The next question follows immediately: does the same axiomatic apparatus reveal load-bearing tensions when applied to Anthropic's specific multi-principle pipeline? I argue it does, and that the tensions are quantifiable.

2. The Trilemma (Master Result). Pin the variant: throughout this section "CAI" denotes the multi-critic variant in which each principle p □ $\{1,\dots,k\}$ carries a separate scalar critic c_p, as opposed to the single-aggregated-critic SL-CAI/RL-CAI of Bai et al. 2022 (arXiv:2212.08073) or the ensemble RLAIF of Lee et al. 2023 (arXiv:2309.00267). The aggregation impossibilities below apply only to the multi-critic variant; the equilibrium claims of section 4 apply to the aggregated and ensemble variants. **Theorem (Trilemma).** Fix $k \geq 3$ principles, $n \geq 3$ critic queries per response, and a finite policy class Π compactified by KL-projection onto a bounded parameter ball. No CAI procedure jointly satisfies all three of {IIA, Pareto, Non-dictatorship} (aggregation leg, Arrow 1951), {DSIC, paraphrase-robustness with quantitative bound $\beta_n \geq k(k+1+\varepsilon)/\varepsilon - 1$ manipulation-distance per Birrell-Pass} (strategic leg, Gibbard 1973, Satterthwaite 1975, Birrell-Pass 2011, with quantitative Gibbard-Satterthwaite refinements via Mossel-Rácz 2014/15 Combinatorica and Isaksson-Kindler-Mossel 2012), and {Bayesian incentive-compatibility under operator-private constitution prior P} (informational leg, asymptotic BNE per Prelec 2004 with robust BTS for $n \geq 3$ per Witkowski-Parkes 2012). *Proof sketch.* Aggregation collapses the k-dimensional principle simplex to a single ranking, exhausting one degree of freedom (Arrow). Strategy-proofness on that ranking forces a near-dictatorial response surface up to manipulation-distance β_n (quantitative GS). BIC under a private prior then forces the residual scoring rule into the proper-scoring family, which the prior-divergence term penalizes (Prelec asymptotic BNE; Witkowski-Parkes BTS, $n \geq 3$). The three reductions exhaust the policy class. Existence of an equilibrium fixed-point on the compactified Π follows from Kakutani applied to the best-response correspondence, where compactness is guaranteed by the KL-projection onto the bounded parameter ball.

3. Sharpened Sub-Result: KL-Regret Bound. The trilemma's informational leg admits a crisp quantitative refinement. **Proposition (R6 T5.1, Constitution-as-Common-Prior).** Let μ° denote the public constitution prior published at training time and $\tilde{\mu}^\circ$ the operator-private deployment prior; let the critic use a strictly proper scoring rule on $n \geq 3$ sampled queries. Then the critic's expected regret relative to the public-prior optimum is sandwiched by

$$\mathrm{KL}(\tilde{\mu}^\circ \| \mu^\circ) - O(\log n / n) \leq \mathrm{Regret}(\text{critic} \mid \tilde{\mu}^\circ) \leq \mathrm{KL}(\tilde{\mu}^\circ \| \mu^\circ) + 2 \cdot \mathrm{TV}(\tilde{\mu}^\circ, \mu^\circ).$$

The lower bound follows from Pinsker plus the asymptotic BNE characterization in Prelec (2004) and the $n \geq 3$ robustness of Witkowski-Parkes (2012); the upper bound from Hoeffding plus a pinwheel coupling argument. The bound supplies a *measurable* proxy for "truth left on the table" by the gap between the published constitution and what the operator actually optimizes for at deployment, in the spirit of Findeis et al.'s Inverse Constitutional AI (2024, arXiv:2406.06560), which already shows constitutions are recoverable from preference data.

---

4. Why This Matters for Anthropic. Constitutional AI is Anthropic's flagship safety mechanism (Bai et al. 2022, arXiv:2212.08073), and its collective extension (Anthropic+ Collective Constitutional AI 2024, arXiv:2406.07814) explicitly invokes pluralist aggregation. OpenAI's Deliberative Alignment (Guan et al. 2024, arXiv:2412.16339) raises the same questions on a different stack. The trilemma converts the open-problems literature on RLHF (Casper et al. 2023, arXiv:2307.15217) and on constitution learnability (Dai-Fleisig 2024, arXiv:2404.13038) from qualitative concern to quantitative price-of-pluralism: any operator running multi-critic CAI must explicitly choose which axiom to relax, and the KL-regret bound prices the third leg in nats. That is exactly the Economics workstream's mandate (preference aggregation, mechanism design, market design for AI deployment), applied to Anthropic's own stack, with mentors Tyler McCrory and Niel Massenkoff positioned to evaluate and sharpen the empirical leg.

5. Empirical Validation ($2300, 8 weeks). I propose one pre-registered headline comparison plus three optional ablations. *Headline.* Frozen-public-constitution critic versus operator-co-trained critic on TruthfulQA-MC1, $n = 4096$ prompts, target effect size $\Delta \geq 4$ percentage points on MC1 accuracy at $\alpha = 0.01$, paired-bootstrap inference, two-sided. Inference cost: 4096 prompts × 2 arms × ~0.05 USD per call ≈ $400. *Optional ablations.* (i) Dispersion-of-principles ablation, varying k □ {3, 5, 9} multi-critic configurations to map the trilemma's k-dependence empirically, $600. (ii) Paraphrase-stability test under DSIC stress, restated accurately as "RLAIF position-bias robustness, Lee et al. 2023 Tab. 4" rather than self-consistency, $300. (iii) Witkowski-Parkes-style peer-prediction self-bootstrap with $n \geq 3$ critic queries to estimate the KL-regret bound's slack term in vivo, $1000. Total: $400 + $600 + $300 + $1000 = $2300. All four protocols pre-registered on OSF before any inference call.

6. Why Me. My unpublished work develops a truth-serum-debate hybrid mechanism that pairs Prelec-style peer-prediction scoring with Irving-Christiano-Amodei (2018, arXiv:1805.00899) debate, whose truth-aligned-equilibrium claim is conjecture supported by a PSPACE/NP analogy rather than theorem, under cooperative-inverse-RL common-knowledge assumptions of Hadfield-Menell et al. (2016, arXiv:1606.03137). The hybrid shows the trilemma's BIC leg can be partly purchased back when debaters share a common prior over the true constitution: peer-prediction Bayesian-truth-serum scoring ($n \geq 3$, asymptotic BNE) restores informational truthfulness at a known KL cost, recoverable from the bound in §3. That is the constructive direction the trilemma's negative result invites; it tells you which leg to give up first and how much it costs in nats. My background in axiomatic decision theory and mechanism design supplies the social-choice machinery (Arrow, Gibbard-Satterthwaite, Mossel-Rácz Combinatorica 2014/15, Isaksson-Kindler-Mossel 2012); my alignment work supplies the CAI-variant pinning the literature so frequently misses. Fellowship outputs over the cohort: a full paper targeted at FAccT, EC, or NeurIPS, plus an internal Anthropic technical memo proposing a *constitution-audit tool* that operationalizes the KL-regret bound as a deployable diagnostic for production CAI pipelines, directly consumable by the Alignment and Trust & Safety teams.

CHANGE LOG

1. **Abstract**: Replaced the em-dash–semicolon construction ("must sacrifice…; I then propose") with a colon and a period. Cut "showing that" (filler).
2. **§1**: Broke the long Conitzer-et-al. sentence (joined by semicolons) into two sentences with named subjects. Replaced "the natural next question is whether" with "The next question follows immediately:

does…” for tighter active voice.

3. **§2**: Replaced the em-dash before “Existence of an equilibrium” with a period (was an LLM-rhythm em-dash, not a genuine aside).
4. **§3**: Replaced em-dash before “which already shows” with a comma (restrictive clause, not aside).
5. **§4**: Replaced em-dash before “OpenAI’s Deliberative Alignment” with a period. Replaced “explicitly invokes pluralist aggregation; OpenAI’s…raises analogous questions” semicolon-chain with two sentences. Replaced “This is the Economics workstream’s mandate — preference aggregation, mechanism design, and market design for AI deployment — applied” with parenthetical phrasing to remove em-dash pair and break the rule-of-three list comma rhythm.
6. **§4**: Cut “directly” (filler intensifier) before “applied to Anthropic’s own stack.”
7. **§6**: Replaced em-dash before “whose truth-aligned-equilibrium claim” with a comma. Replaced em-dash before “directly consumable” with a comma.
8. **§6**: Replaced “This is exactly the constructive direction the trilemma’s negative result invites — it says which leg” with “That is the constructive direction the trilemma’s negative result invites; it tells you which leg” (removed em-dash, removed “exactly” intensifier, changed “says” to “tells you” for active register).
9. **§6**: Cut “and how much it costs in nats” was preserved (technical), but the surrounding “exactly the constructive direction” lost its “exactly.”
10. **Global**: Verified preservation of every theorem statement, equation (KL/TV bound), citation (all arXiv IDs, years, page refs), defined term (CAI, BIC, DSIC, BTS, IIA, Pareto, KL, TV, BNE, RLHF, RLAIF, GS, FAccT, EC, NeurIPS), section header, and the budget arithmetic (\$400 + \$600 + \$300 + \$1000 = \$2300, $n = 4096$, $\Delta \geq 4$pp, $\alpha = 0.01$). No technical content altered.